

STUDENT COURSE GUIDE

Applied AI in Business

From useful questions to reliable organizational capability

8 sessions | 4 hours each | 32 contact hours

Course promise: You will learn to identify a valuable use of AI, build and test a practical prototype, redesign the surrounding work, and defend whether an organization should fund, pilot, redesign, or stop it.

For students of business, management, information systems, analytics, and related fields

Contents

1 Welcome	1
2 What This Course Is About	1
2.1 Central course question	1
3 Format and Expectations	1
3.1 Course information	2
4 Learning Outcomes	2
5 Materials and Tools	3
5.1 What students need	3
5.2 What the course provides	3
5.3 Readiness before Session 1	3
6 Responsible Use of AI in the Course	3
7 Course Map	4
8 Assessment	4
Session 1: AI Is Changing Business	6
Session 2: Working Effectively with Generative AI	7
Session 3: From AI Idea to Business Value	8
Session 4: Data, Knowledge, and Predictive AI	9
Session 5: Classification, Scoring, and Prioritization	10
Session 7: Evaluating and Improving AI Systems	11
Session 7: LLMs for Business Analysis and Communication	12
Session 8: Scaling AI and Organizational Transformation	13
9 Capstone Project	14
9.1 Minimum requirements	14
9.2 Permitted scope	14
10 Final Capstone Rubric	14
11 Student FAQ	15
12 How to Approach Open-Ended Work	16
13 Working Principles	16

14 Literature and Reading Guide	16
14.1 Core course literature	17
14.2 Reading pathway by session	17
14.3 Additional perspectives	18

1. Welcome

This course will be different. We will not treat artificial intelligence as a collection of fashionable tools, and we will not pretend that business problems arrive as tidy exercises with one correct answer. We will begin with real organizational work: decisions, customers, documents, delays, mistakes, incentives, and people who may reasonably disagree.

AI has already changed the role of the business analyst. Models can draft requirements, summarize meetings, explore files, generate SQL, prepare reports, and produce a surprisingly confident slide deck before lunch. The role is not disappearing, but its centre of gravity is moving. The valuable analyst increasingly frames the problem, supplies context, challenges evidence, redesigns work, evaluates failures, and decides where human judgment must remain.

This is therefore a course about using AI in business, not merely talking about it. You will work extensively with generative AI and may use AI-assisted development tools. You do not need to be an experienced programmer. Some knowledge of Python will help, but good judgment, curiosity, and the ability to ask precise questions will matter more.

Expect open-ended problems. There may be several defensible answers, but there are still weak assumptions, misleading evidence, irresponsible designs, and expensive mistakes. “There is no single correct answer” does not mean “anything goes.”

A useful habit: Before asking what the AI can produce, ask what decision or process should improve, for whom, and how anyone will know.

2. What This Course Is About

The course follows the lifecycle of an AI-enabled business capability:

Stage	Central question
Understand	What can different forms of AI actually do, and where do they fail?
Select	Which business problem is worth solving, and what value could change?
Ground	What data, documents, context, and organizational knowledge are required?
Redesign	How should people, rules, models, and automation work together?
Evaluate	Does the system work reliably on realistic and difficult cases?
Govern	Who owns the outcome, what may the system access, and how is harm controlled?
Scale	Can the prototype become an affordable, adopted, and sustainable capability?

Traditional business analytics remains in the course where it is essential: defining outcomes, checking data, interpreting predictions, establishing baselines, and measuring impact. It is a supporting language, not the plot of the entire film.

2.1 Central course question

Where should an organization use AI, what must change for it to create value, and what evidence would justify trusting and scaling it?

3. Format and Expectations

3.1 Course information

Item	Information
Lecturer and contact	Armando Vieira, armando.vieira@ut.ee
Learning platform	Moodle
Online access	https://ut-ee.zoom.us/j/94825400838?pwd=CUGn8q31rKx18Jkqa8Ao01Lti71192.1 passcode 189259
Other resources	https://github.com/farquasar1/AI_Business

Format: Eight sessions of four contact hours. Each contains approximately two hours of concepts, cases, and demonstrations followed by a two-hour practical workshop. Independent capstone work is expected between sessions.

Teaching approach: Short explanations, concrete business cases, demonstrations, team construction, peer critique, failure analysis, and revision. We will spend less time admiring definitions and more time discovering where they break.

Audience: Advanced undergraduate, master's, executive-education, and professional students who want to evaluate, design, or implement AI in organizational settings.

Prerequisites: Basic spreadsheet literacy and willingness to reason with evidence. Basic statistics are useful. Python is helpful but not compulsory if the initial setup activities are completed.

Student responsibility: You remain responsible for every submitted claim, number, recommendation, and line of generated code. "The model said so" is not a method, a citation, or a career strategy.

4. Learning Outcomes

After completing the course, you should be able to:

- Explain the practical capabilities and limitations of predictive, generative, and automation technologies.
- Translate an organizational problem into a user, decision, workflow, baseline, value hypothesis, and measurable outcome.
- Prioritize AI opportunities using value, feasibility, readiness, risk, and strategic relevance.
- Recognize opportunities for AI-enabled products, services, customer experiences, and business models—not only internal efficiency.
- Design effective prompts and context, including structured outputs and evidence-grounded use of organizational knowledge.
- Compare general, specialized, large, small, cloud, local, proprietary, and open models using quality, latency, privacy, cost, and dependence.
- Identify how text, documents, speech, images, video, interfaces, and structured data can be combined in multimodal business systems.
- Decide what should be eliminated, simplified, assisted, automated, or retained under human control.
- Design human–AI interactions that support questioning, correction, evidence inspection, uncertainty, escalation, and meaningful intervention.
- Build and test a practical AI-enabled prototype using low-code or code-based tools.
- Evaluate AI outputs using realistic task sets, business consequences, failure categories, and appropriate human review.
- Design a feedback and organizational-learning loop that keeps knowledge, rules, tests, and workflows useful after deployment.
- Address interpretability, privacy, security, permissions, accountability, and auditability as design requirements.
- Estimate costs, adoption needs, integration effort, and expected organizational benefits.
- Present and defend a fund, pilot, redesign, or stop recommendation to decision makers.

5. Materials and Tools

The course is vendor-neutral. Specific tools are used because they make practical work easier, not because their logos are learning outcomes.

5.1 What students need

Item	Purpose and expectation
Laptop and browser	Access course materials, AI systems, notebooks, and workflow tools.
Approved LLM access	Analyse text and documents, generate structured outputs, critique assumptions, and prototype interactions. You don't need to use the latest models. Deepseek, Gemini flash or GLM-5 are good enough
AI-assisted workspace	Claude code, Cursor or ChatGPT (codex) are recommended for understanding, generating, testing, and improving code and project files.
Notebook environment	Jupyter or Google Colab for lightweight data inspection and evaluation.
Python, optional but useful	Python 3.11+, pandas, matplotlib, scikit-learn, and pytest for students choosing a code-based path.
Workflow environment	n8n or an equivalent visual tool for connecting triggers, information, AI steps, approvals, and actions.
Business tools	Spreadsheet, collaborative whiteboard, and presentation software.
Version control	Git or an institution-approved alternative for reproducible project work.

5.2 What the course provides

- A coherent business case, privacy-safe datasets, and a collection of policies, reports, product information, and customer messages.
- Starter notebooks, sample workflows, schemas, tests, and step-by-step setup instructions.
- Opportunity, workflow, prompt, evaluation, risk, cost, and pilot templates.
- Mock organizational systems and APIs suitable for safe experimentation.
- Normal, ambiguous, incomplete, adversarial, and failure-recovery test scenarios.

5.3 Readiness before Session 1

- Confirm access to the approved LLM and course workspace.
- Install Cursor and Git if you intend to use the code path.
- Confirm that Jupyter or Google Colab opens the supplied notebook.
- Never place passwords, API keys, personal data, or confidential organizational material in prompts or source control.
- Bring one recurring business task or decision that is frustrating, slow, costly, or unusually dependent on expert knowledge.

Two valid paths: Students may use low-code tools or Python. Assessment focuses on the quality of the problem framing, system design, evidence, testing, and business recommendation—not on producing the largest amount of code.

6. Responsible Use of AI in the Course

AI use is expected - in fact mandatory - not hidden. You may use it to explore ideas, interpret documents, generate or explain code, create test cases, critique drafts, and improve communication. However, each submission must include a short AI-use record stating:

- which tools were used and for what purpose;
- what important suggestion was accepted, modified, or rejected;
- what was independently checked and how;
- what limitations or uncertainties remain.

You must not submit confidential data to unapproved services, fabricate sources, conceal material AI use, or present generated output that you cannot explain. Productive AI use reduces mechanical work while increasing the standard of verification.

7. Course Map

No.	Session	Capability developed
1	AI Is Changing Business	Recognize meaningful AI opportunities and connect them to real work, decisions, and organizational consequences.
2	Working Effectively with Generative AI	Design context and structured interactions; understand multimodality and select models using explicit trade-offs.
3	From AI Idea to Business Value	Evaluate process improvement, new products, services, and business models through baselines, value, costs, and pilots.
4	Data, Knowledge, and Predictive AI	Combine structured data, organizational knowledge, multimodal information, retrieval, and only necessary predictive components.
5	Human–AI Workflows and Process Redesign	Design work, interfaces, correction, evidence, and responsibility across people, rules, models, and automation.
6	Evaluating and Improving AI Systems	Test quality, reliability, interpretability, cost, and consequences; design feedback and improvement loops.
7	Governance, Security, and Responsible Deployment	Design permissions, accountability, safeguards, auditability, and recovery proportional to risk.
8	Scaling AI and Organizational Transformation	Turn a prototype into an adopted capability and defend a fund, pilot, redesign, or stop decision.

8. Assessment

The components below indicate the intended evidence, but their weights, individual/team allocation, deadlines, and submission details are deliberately left open.

Provisional component	Evidence and status
AI opportunity and value brief	Business problem, user, workflow or product opportunity, baseline, expected value, assumptions, and prioritization.
Generative AI and knowledge task	Context specification, structured or multimodal output, model comparison, evidence checks, and failure analysis.
Human–AI workflow milestone	Redesigned process, interaction design, working prototype, human decisions, controls, logs, and simpler alternative.
Evaluation and governance review	Test set, results, consequences, interpretability, feedback loop, risk register, and deployment decision.
Final capstone	Integrated prototype, business case, demonstration, economics, governance, operating model, and roadmap. team size of 2, max 3, and individual evidence
Professional practice	Preparation, peer critique, reproducibility, AI-use record, reflection, and constructive teamwork.

Assessment principle: A polished demonstration cannot compensate for an irrelevant problem or weak evidence. Finding a serious failure, simplifying the design, or recommending that a project stop can be excellent work when the reasoning is strong.

SESSION 1**AI Is Changing Business**

4 hours | 2 hours concepts + 2 hours practical

Understanding the new capability landscape and finding business problems worth addressing.

Questions for the session

- What changed with generative AI, and what did not?
- Which parts of professional and managerial work are being automated, augmented, or redesigned?
- Where can AI improve cost, speed, quality, customer experience, learning, or coordination?
- How is the role of the business analyst changing from producing documents to curating context and value?
- How can we distinguish a valuable opportunity from AI theatre?

Concept block – 120 minutes

1. 0–25 min – Course framing and the changing role of the business analyst.
2. 25–55 min – Predictive, generative, optimization, and automation capabilities through business examples.
3. 55–80 min – Overview of AI capabilities.
4. 80–90 min – Tasks, decisions, processes, and organizational capabilities.
5. 90–110 min – Opportunity criteria: value, feasibility, readiness, risk, and strategic relevance.
6. 110–120 min – From an interesting idea to a question worth investigating.

Practical block – 120 minutes

1. 120–150 min – Warm-up: an overview of coding and use of AI tools'
2. 150–175 min – Team case: define the real problem, decision, owner, and evidence required.
3. 185–215 min – Data-request auction under a limited information budget.
4. 215–235 min – Teams stress-test each other's assumptions and metrics..
5. 235–240 min – Individual exit reflection.

Practical output: One-page AI opportunity brief containing the problem, user, decision, present workflow, expected benefit, necessary evidence, principal risk, and one question that could change the recommendation.

Tools and materials: A business case to be presented; opportunity canvas; limited-budget data cards; approved LLM for critique; collaborative whiteboard or spreadsheet.

SESSION 2

Working Effectively with Generative AI

4 hours | 2 hours concepts + 2 hours practical

Treating prompting as context and task design rather than conversational magic.

Questions for the session

- What are tokens, context windows, prompts, and structured outputs?
- Why is a context window not the same as memory or understanding?
- How do text, speech, images, documents, video, and interface context change the task?
- How should a complex business task be decomposed into checkable steps?
- What changes when the same problem is viewed by a customer, manager, analyst, or regulator?
- When is an LLM the wrong tool?

Concept block – 120 minutes

1. 0–30 min – A practical mental model of large language models.
2. 30–55 min – Tokens, context-window limits, attention, context versus memory, and non-determinism.
3. 55–65 min – Break.
4. 65–85 min – How to design prompts, goals, loops, context, examples, constraints, structured outputs, and abstention.
5. 85–102 min – Multimodal AI: speech, images, documents, screens, and structured data.
6. 102–120 min – Model-selection literacy: quality, latency, privacy, cost, openness, and when to use rules, databases, or people.

Practical block – 120 minutes

1. 120–145 min – Compare weak, improved, and structured context specifications.
2. 145–180 min – Analyse one case through four organizational roles and two information modalities.
3. 180–190 min – Break.
4. 190–220 min – Create and test an output schema with explicit evidence and uncertainty fields.
5. 220–240 min – Compare two model or tool choices, red-team the design, and revise it.

Practical output: A reusable context specification, structured or multimodal output schema, model comparison, and short account of what was verified, rejected, or left uncertain.

Tools and materials: Approved text and multimodal AI tools; prompt/context template; JSON or table schema; role cards; model-choice matrix; adversarial and incomplete-information cases.

SESSION 3**From AI Idea to Business Value**

4 hours | 2 hours concepts + 2 hours practical

Connecting an AI capability to a baseline, an outcome, and a credible investment decision.

Questions for the session

- What changes if the system works, and compared with what?
- Does AI improve an existing process, or enable a new product, service, customer experience, or revenue model?
- How do time savings become real value rather than an attractive multiplication exercise?
- Which costs and adoption barriers are commonly omitted from AI business cases?
- Should the organization build, buy, partner, or first learn through a small pilot?
- What evidence would justify stopping?

Concept block – 120 minutes

1. 0–30 min – User, action, outcome, counterfactual, and baseline.
2. 30–55 min – Value mechanisms: revenue, margin, cost, quality, speed, risk, and experience.
3. 55–65 min – Break.
4. 65–85 min – AI-enabled products and business models: personalization, expertise at scale, adaptive services, outcome pricing, and new distribution.
5. 85–102 min – Total cost of ownership, model choice, inference economics, integration, review, and maintenance.
6. 102–120 min – Build, buy, partner, staged investment, learning milestones, and stop conditions.

Practical block – 120 minutes

1. 120–145 min – Compare a process-improvement concept with a new product or service concept.
2. 145–180 min – Construct a baseline and value tree; estimate benefits, operating costs, and sensitivities.
3. 180–190 min – Break.
4. 190–220 min – Compare model, build, buy, and partner options.
5. 220–240 min – Present a pilot, redesign, or stop recommendation to a sceptical CFO.

Practical output: Comparison of efficiency and innovation opportunities, value tree, baseline or market alternative, cost model, key assumptions, model and sourcing comparison, pilot criteria, and stop conditions.

Tools and materials: Spreadsheet; value and cost templates; vendor scenario cards; approved LLM as assumption challenger.

SESSION 4**Data, Knowledge, and Predictive AI**

4 hours | 2 hours concepts + 2 hours practical

Providing the evidence and organizational context that useful AI systems require.

Questions for the session

- Which information belongs in a database, a document collection, a predictive model, or a business rule?
- How should speech, images, documents, interface events, and structured records be connected?
- What is the difference between retrieval, fine-tuning, and supplying more context?
- When are forecasting, scoring, classification, or segmentation genuinely useful?
- How do data quality, provenance, permissions, and freshness affect system reliability?
- What can be concluded from a prediction, and what remains a business decision?

Concept block – 120 minutes

1. 0–25 min – Structured, unstructured, and multimodal organizational information.
2. 25–50 min – Retrieval-augmented generation, embeddings, multimodal search, grounding, and source boundaries.
3. 50–60 min – Break.
4. 60–100 min – Essential predictive AI: forecasting, scoring, classification, segmentation, and recommendation.
5. 100–120 min – Baselines, leakage, uncertainty, correlation, and causal overclaims.

Practical block – 120 minutes

1. 120–145 min – Map the structured, document, image, audio, and human sources required by the selected use case.
2. 145–180 min – Build a small retrieval, extraction, or predictive component using Python.
3. 180–190 min – Break.
4. 190–220 min – Test missing, outdated, conflicting, and access-restricted information.
5. 220–240 min – Decide whether the component is useful, unnecessary, or requires more evidence.

Practical output: Multimodal information architecture, source inventory, prototype component, test results, and a clear account of what the system knows, does not know, and should not access.

Tools and materials: Course datasets and documents; approved LLM; notebook or low-code data tool; retrieval toolkit where appropriate; data dictionary and provenance template.

SESSION 5

Classification, Scoring, and Prioritization

4 hours | 2 hours concepts + 2 hours practical

Building scoring systems for churn, risk, fraud, or sales while choosing thresholds based on value.

Learning objectives

- Train interpretable and ensemble classification models.
- Evaluate ranking and classification quality with suitable metrics.
- Choose thresholds using capacity, intervention cost, and error asymmetry.
- Check calibration, segment performance, fairness, and stability.

Core concepts

- Logistic models, decision trees, ensemble methods, probability scores, and feature importance.
- Confusion matrices, precision, recall, ROC-AUC, PR-AUC, lift, gain, and calibration.
- Threshold optimization and constrained prioritization under limited staff capacity.
- Data leakage, class imbalance, proxy variables, drift, and responsible use.

Session plan

1. Theory (0–120 min)

- 0–40 min — From probability to intervention: why classification is not the final decision.
- 40–80 min — Models and evaluation metrics.
- 80–95 min — Break.
- 95–120 min — Thresholds, calibration, and operational constraints.

2. Practice (120–240 min)

- 120–200 min — Lab: develop a churn or lead-priority score.
- 200–225 min — Red-team review of leakage, bias, and unstable features.
- 225–240 min — Decision policy submission.

Hands-on challenge

Build a customer score, compare at least two models, select an action threshold for a fixed intervention budget, and estimate incremental business value. **Required evidence:** Scoring pipeline, evaluation dashboard, threshold rationale, and brief risk assessment. **Preparation:** Review probability, sampling, and confusion matrices. **Tools and materials:** Python; scikit-learn; calibration and model-explanation utilities; a notebook or lightweight dashboard.

SESSION 7**Evaluating and Improving AI Systems**

4 hours | 2 hours concepts + 2 hours practical

Replacing “it looked good in the demo” with evidence, feedback, and controlled improvement.

Questions for the session

- What does good performance mean for this specific task and user?
- How should open-ended outputs be evaluated?
- When is an interpretable, simpler system preferable?
- How do model errors translate into financial, operational, legal, or human consequences?
- What should be monitored after deployment?
- Which corrections should become new tests, knowledge, rules, prompts, or model changes?

Concept block – 120 minutes

1. 0–30 min – Model performance, task performance, and business performance.
2. 30–55 min – Evaluation sets, rubrics, expert review, comparison, and uncertainty.
3. 55–65 min – Break.
4. 65–85 min – Interpretability, individual explanations, business justification, and false confidence.
5. 85–103 min – Edge cases, adversarial testing, drift, and failure taxonomies.
6. 103–120 min – Learning loops: monitoring, expert corrections, test expansion, knowledge updates, versioning, and rollback.

Practical block – 120 minutes

1. 120–145 min – Define acceptance criteria and construct a balanced evaluation set.
2. 145–180 min – Test the prototype and record evidence, uncertainty, latency, and cost.
3. 180–190 min – Break.
4. 190–215 min – Red-team plausible but wrong outputs and high-consequence cases.
5. 215–230 min – Convert selected failures and expert corrections into durable tests or knowledge updates.
6. 230–240 min – Recommend deploy, restrict, redesign, collect more evidence, or stop.

Practical output: Evaluation plan, test set, results table, failure taxonomy, business-consequence analysis, interpretability requirement, feedback and improvement loop, and deployment recommendation.

Tools and materials: Scenario dataset; evaluation rubric; logs; spreadsheet or notebook; testing utilities; cost and latency tracker.

SESSION 7

LLMs for Business Analysis and Communication

4 hours | 2 hours concepts + 2 hours practical

Using LLMs for evidence-grounded analysis, structured outputs, visualization explanation, and stakeholder communication.

Learning objectives

- Select LLM tasks based on input, output, risk, and verification requirements.
- Design prompts with context, examples, constraints, and structured outputs.
- Build an evaluation set and measure task success beyond subjective impressions.
- Turn evidence into accurate narratives tailored to operational and executive audiences.

Core concepts

- Tokens, context, model behavior, structured generation, and prompt patterns.
- Extraction, classification, synthesis, chart explanation, drafting, and natural-language interfaces.
- Retrieval-augmented generation: chunking, search, grounding, citations, and access control.
- Evaluation, hallucination, abstention, narrative structure, audience adaptation, and uncertainty communication.

Session plan

1. Theory (0–120 min)

- 0–35 min — Choosing high-value LLM tasks and rejecting weak ones.
- 35–75 min — Prompt and structured-output design.
- 75–90 min — Break.
- 90–120 min — Retrieval, grounding, and evaluation beyond “it looks good”.

2. Practice (120–240 min)

- 120–195 min — Lab: document-to-structured-data and evidence-backed report generation.
- 195–225 min — Evaluate and rewrite the result for operational and executive audiences.
- 225–240 min — Failure log submission.

Hands-on challenge

Process business documents and analytical outputs, extract fields into a validated schema, retrieve supporting evidence, explain one visualization, and generate audience-specific management reports. **Required evidence:** Prompt or pipeline specification, evaluation dataset, results table, failure taxonomy, operational summary, and executive narrative. **Preparation:** Collect or use the supplied non-sensitive documents and define the exact output schema before class. **Tools and materials:** Institution-approved LLM API or local model; JSON Schema or Pydantic; document parser; retrieval toolkit; visualization and presentation tools.

SESSION 8**Scaling AI and Organizational Transformation**

4 hours | 2 hours concepts + 2 hours practical

Turning a promising prototype into an adopted, affordable, and sustainable capability.

Questions for the session

- What separates a prototype from a production capability?
- Can the system scale economically, technically, and organizationally?
- How will roles, skills, incentives, and trust change?
- Who maintains the data, knowledge, tests, workflow, and operating decisions?
- How will the organization preserve useful corrections and learn across teams and use cases?
- What is the next justified investment—fund, pilot, redesign, or stop?

Concept block – 120 minutes

1. 0–25 min – Prototype versus production: reliability, latency, integration, and cost.
2. 25–50 min – Scaling from one user to an organization; reusable capabilities, new services, and isolated applications.
3. 50–60 min – Break.
4. 60–80 min – Operating models, ownership, vendor dependence, and maintenance.
5. 80–100 min – Adoption, trust, incentives, role redesign, preserving expertise, and organizational learning loops.
6. 100–120 min – Executive communication, staged investment, benefits realization, and portfolio learning.

Practical block – 120 minutes

1. 120–190 min – Team executive presentations and live demonstrations.
2. 190–200 min – Break.
3. 200–225 min – Injected ambiguity, failure, permission, or scale scenario.
4. 225–235 min – Executive panel decision: fund, pilot, redesign, or stop.
5. 235–240 min – Individual reflection: what changed in the recommendation and why?

Practical output: Final prototype, executive narrative, evaluation evidence, cost and value case, governance design, learning loop, operating model, and staged implementation roadmap.

Tools and materials: Stable demonstration environment; version-controlled repository or documented low-code workflow; presentation tools; logs; cost model; panel rubric.

9. Capstone Project

The capstone is an AI-enabled business capability, not merely a chatbot, model, or collection of screenshots. It may redesign an existing process or propose an AI-enabled product, service, customer experience, or business model in marketing, sales, finance, operations, customer service, knowledge work, human resources, healthcare, public administration, or another approved context.

9.1 Minimum requirements

The final submission must include:

- A concrete problem or opportunity, target user, decision or product owner, present baseline or credible alternative, and value metric.
- A justified AI capability and a credible comparison with a simpler non-AI alternative.
- The data, documents, context, rules, and organizational knowledge required by the system.
- A redesigned workflow showing what people, AI, and deterministic systems each do.
- At least one working AI-enabled step with defined inputs, outputs, and acceptance criteria.
- A realistic evaluation set including normal, incomplete, ambiguous, and high-consequence cases.
- A feedback and improvement loop explaining how corrections become updated tests, knowledge, rules, prompts, workflows, or model decisions.
- Explicit human decision points, permissions, escalation, logs, and recovery.
- An economic case covering expected benefit, operating cost, adoption effort, and important assumptions.
- A build, buy, or partner position and staged pilot-to-scale roadmap.
- An AI-use record and a candid account of remaining uncertainty.

9.2 Permitted scope

The project may be implemented with low-code tools, Python, or a combination. A bounded agent is optional and should appear only when it produces a clear benefit over a deterministic workflow. Larger technical scope does not automatically produce a stronger project.

10. Final Capstone Rubric

The criteria below describe the intended direction of assessment.

Provisional criterion	Strong evidence
Problem, innovation, and value	The problem or opportunity, user, owner, alternative, value mechanism, assumptions, and consequences are precise and credible.
AI, model, and context design	The capability, modality, model choice, information, context, structured outputs, and simpler alternatives are understood and justified.
Workflow and human experience	Work allocation, interaction, evidence, uncertainty, correction, exceptions, and responsibility form a coherent design.
Evaluation and learning	Tests are realistic and reproducible; failures, uncertainty, interpretability, cost, consequences, feedback, and improvement are analysed honestly.
Governance and implementation	Permissions, privacy, security, accountability, economics, adoption, ownership, learning, and roadmap are proportionate and practical.
Communication	The demonstration and executive case are clear, concise, evidence-based, and responsive to difficult questions.

11. Student FAQ

Do I need to know how to code?

No. You may follow a low-code route, a Python route, or combine them. Basic coding makes some prototypes easier, but assessment will focus on problem framing, system design, evidence, testing, and business consequences. You must still be able to explain anything generated on your behalf.

Are we expected to use AI?

Yes. AI use is part of the course. The objective is not to hide it but to use it critically and document important choices.

Must I submit prompts or AI conversations?

You must maintain a concise AI-use record.

[TO BE CONFIRMED: whether full prompts, selected conversations, tool logs, or only material interactions must be submitted]

Will assignments be individual or completed in teams?

The course will include collaborative practical work and individual accountability. Max Team size of 2

How will open-ended answers be graded?

Marks will reward clear assumptions, relevant evidence, recognition of uncertainty, consequences, and the ability to revise a position.

Can a project receive a strong grade if the recommendation is to stop?

Yes. A well-supported no-go recommendation can be more valuable than an enthusiastic prototype built around a weak problem. The quality of the reasoning and evidence matters more than whether the conclusion is optimistic.

Can I use a problem from my workplace?

Potentially, but only with appropriate permission and privacy protection. Do not upload confidential, personal, regulated, or commercially sensitive information to unapproved tools.

What must the final project contain?

TBC

What are the deadlines and attendance requirements?

TBC

What happens if work is late or needs revision?

Not acceptable

What if the recommended tools change during the semester?

The course is vendor-neutral. Tools may change when access, quality, price, or institutional policy changes. Learning outcomes concern transferable practices: context design, model selection, workflow design, evaluation, governance, and value.

Where can I ask for help or accessibility arrangements?

Moodle open forum. Keep discussions open to all

12. How to Approach Open-Ended Work

For most assignments, use the following sequence:

1. **Frame:** Who faces what problem, and which decision or action must improve?
2. **Question:** What would we need to know before acting?
3. **Baseline:** How is the work performed now, with what cost and failure rate?
4. **Design:** What should AI, software rules, and people each do?
5. **Test:** On which ordinary and difficult cases could the design succeed or fail?
6. **Interpret:** What do the results mean for the business rather than only for the model?
7. **Decide:** Fund, pilot, redesign, collect more evidence, or stop?

Good questions are part of the assessed work. A strong question exposes an assumption, changes the evidence required, or reveals that the apparent problem is not the real one.

13. Working Principles

Principle	Meaning
Problem before product	Begin with the business situation, not with a preferred model or vendor.
Decision before output	Define what someone will do differently if the system succeeds.
Context before cleverness	Relevant definitions, evidence, constraints, and examples usually matter more than verbal flourish.
Baseline before sophistication	Compare with the present process and a simple alternative.
Redesign before automation	Remove unnecessary work before accelerating it.
Evidence before demonstration	Test repeatable performance and failure behaviour, not one curated interaction.
Human agency by design	Make judgment, recourse, responsibility, and the capacity to intervene real rather than ceremonial.
Governance throughout	Privacy, security, permissions, and accountability shape the architecture from the beginning.
Economics and adoption matter	A capable system creates no value when nobody trusts, uses, maintains, or can afford it.
Questions create options	The best analyst does not merely answer faster; they improve what the organization is able to ask and decide.

Final thought: The objective is not to become the person who can make AI produce the most material. It is to become the person who can make an organization ask better questions, redesign work intelligently, and act on evidence without surrendering judgment.

14. Literature and Reading Guide

The literature combines business strategy, empirical research, system design, and governance. It is deliberately selective: the aim is to develop a useful intellectual toolkit, not to produce a heroic unread pile of PDFs.

Readings are marked as follows:

- **Core** — an anchor source used across several sessions;
- **Prepare** — the principal reading or extract for a particular session;
- **Extend** — optional material for deeper technical, strategic, or governance study;
- **Reference** — a resource to consult when designing or evaluating a project.

Specific chapters, extracts, and any access arrangements will be announced before each session. Because the field changes rapidly, the lecturer may replace a reading with a newer or more relevant source while preserving the same learning purpose.

14.1 Core course literature

1. **Mollick, E. (2024). *Co-Intelligence: Living and Working with AI*. Portfolio.**
A practical introduction to working with generative AI, understanding its uneven capabilities, and redesigning knowledge work without confusing fluency with truth.
2. **Agrawal, A., Gans, J., & Goldfarb, A. (2022). *Power and Prediction: The Disruptive Economics of Artificial Intelligence*. Harvard Business Review Press.**
A strategy lens for understanding AI as cheaper prediction and for identifying the complementary decisions, actions, data, and workflow changes required to create value.
3. **Iansiti, M., & Lakhani, K. R. (2020). *Competing in the Age of AI*. Harvard Business Review Press.**
A useful account of how AI changes operating models, scale, learning, and the architecture of firms rather than merely improving isolated tasks.
4. **Stanford Institute for Human-Centered AI (2026). *AI Index Report 2026*.**
A current evidence base on capabilities, adoption, economics, responsible AI, policy, and public attitudes. Students should use relevant chapters rather than attempt the traditional academic sport of reading every page.
5. **National Institute of Standards and Technology (2024). *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile (NIST AI 600-1)*.**
A practical reference for identifying, measuring, managing, and governing risks specific to generative AI.

14.2 Reading pathway by session

Session	Prepare	Extend or consult
1	Stanford HAI, <i>AI Index Report 2026</i> : selected sections on capability, adoption, and the economy.	Brynjolfsson, E., Li, D., & Raymond, L. R. (2025). " Generative AI at Work ." <i>Quarterly Journal of Economics</i> . Focus on what the field experiment measures—and what it does not.
2	Mollick, <i>Co-Intelligence</i> : selected chapters on working with AI and the "jagged frontier."	Liu, N. F. et al. (2024). " Lost in the Middle: How Language Models Use Long Contexts ." <i>Transactions of the ACL</i> , 12, 157–173.
3	Agrawal, Gans, & Goldfarb, <i>Power and Prediction</i> : selected chapters on prediction, decisions, and system redesign.	Iansiti & Lakhani, <i>Competing in the Age of AI</i> : selected chapters on strategy and operating models.
4	Lewis, P. et al. (2020). " Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks ." <i>NeurIPS 2020</i> . Read for the central design idea, not every equation.	Noy, S., & Zhang, W. (2023). "Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence." <i>Science</i> , 381(6654), 187–192.
5	Amershi, S. et al. (2019). " Guidelines for Human–AI Interaction ." <i>CHI 2019</i> .	Dell'Acqua, F. et al. (2023). " Navigating the Jagged Technological Frontier ." Consider the "centaur" and "cyborg" patterns of human–AI work.
6	Dell'Acqua et al., "Navigating the Jagged Technological Frontier": results, boundary conditions, and implications for evaluation.	NIST AI 600-1: selected sections on measurement, testing, incident disclosure, and human oversight.
7	NIST AI 600-1: selected risk categories and recommended actions.	European Union (2024). Regulation (EU) 2024/1689, the Artificial Intelligence Act : scope, risk categories, prohibited practices, transparency, and organizational obligations.

Session	Prepare	Extend or consult
8	Iansiti & Lakhani, <i>Competing in the Age of AI</i> : selected chapters on scale, architecture, and organizational learning.	Sculley, D. et al. (2015). " Hidden Technical Debt in Machine Learning Systems ." <i>NIPS 2015</i> . A reminder that a successful demo can be the first payment on a very expensive loan.

14.3 Additional perspectives

These sources are useful for essays, capstone projects, or students who want a stronger management and organizational lens:

- Raisch, S., & Krakowski, S. (2021). "Artificial Intelligence and Management: The Automation–Augmentation Paradox." *Academy of Management Review*, 46(1), 192–210.
- Davenport, T. H., & Mittal, N. (2022). *All-in on AI: How Smart Companies Win Big with Artificial Intelligence*. Harvard Business Review Press.
- Students should also consult current documentation for the models, platforms, and data sources used in their work. Vendor claims are inputs to evaluation, not substitutes for it.

How to read for this course: For each source, ask: What claim is being made? What evidence supports it? Under what conditions might it fail? What decision would change if the claim were true? A highlighted PDF is not yet an analysis.